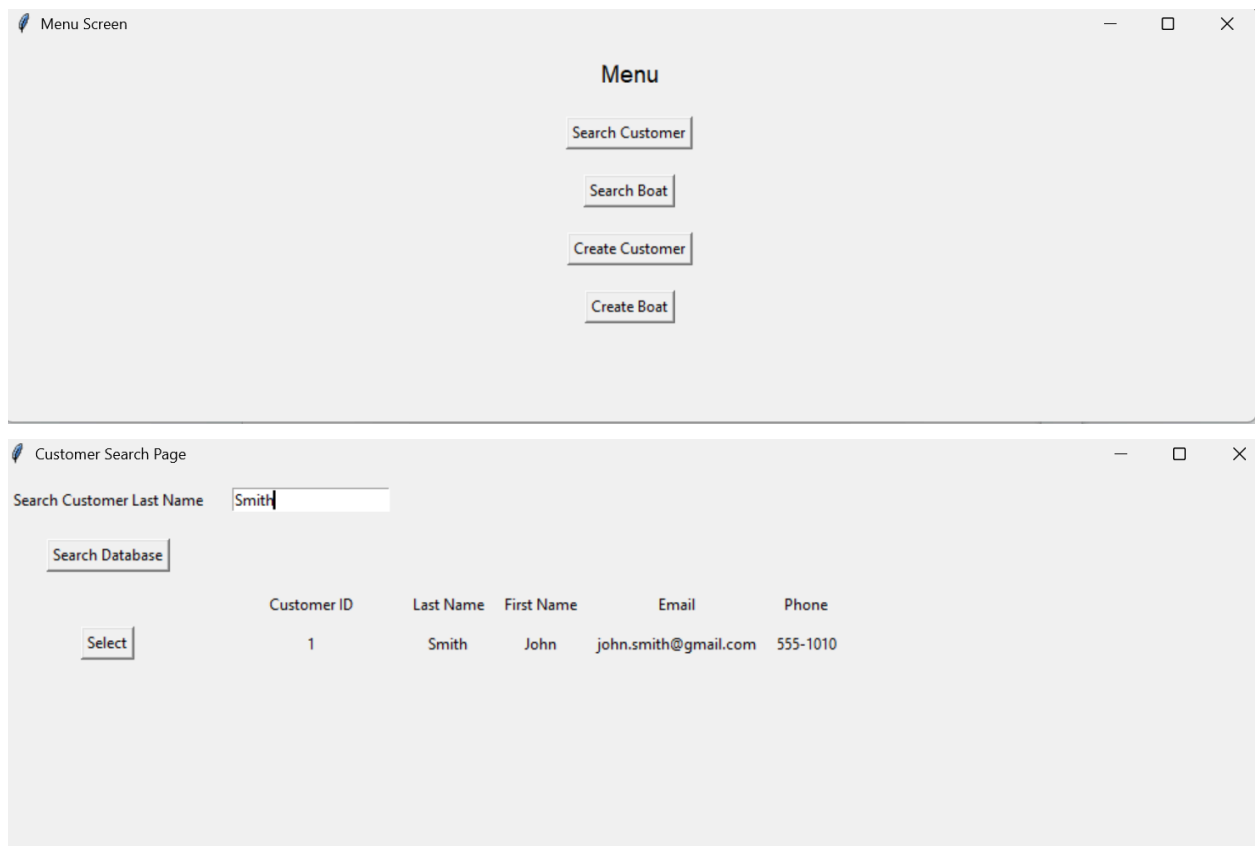


System Report/Overview

My Program is a simple Reservation Manager for a small boat harbor. It has a MySQL Database and a tkinter gui. The four tables in the database are **reservations**, **customers**, **boats** and **slips**. I used Tkinter to make a menu page that allows the user to navigate to separate windows for four different functions. The four functions are Search Customer, Search Boat, Create Customer, and Create Boat. Reservations can be made on a customer's page. I did not include a function to create slips because regular harbor staff would never do that. That would happen when the harbor is built and the software is installed. By searching by boat or customer you can navigate to a customer page that allows you to edit customer information, see the boats and reservations attached to that customer, and create new reservations. To create a new reservation you enter the length of the boat that you need and the search function will return a list of slips that meet the length requirement. When you select a slip it will automatically be entered in the slip number field that is used to create a reservation. This is the main problem that needed solving. Now to find a viable slip you just need to enter in the length and select a slip. If you don't want to install something to use the database I would be happy to stop by for office hours and show you on my computer to prove that it works.

Testing Screen Shots:



Customer Page

Existing Customer

Last Name

Smith

First Name

John

Email

john.smith@gmail.com

Phone

555-1010

Save Edit

Boats

BT101A

28

BT202B

32

Reservations

1

1

BT101A

10

2025-04-01

2025-04-15

2

1

BT202B

11

2025-05-10

2025-05-18

7

1

BT202B

14

1234-02-01

1234-03-05

Create Reservation

Boat ID

BT202B

Start Date

2025-04-01

End Date

2025-05-03

Enter Length

25

Slip Number

4

Search

Save Reservation

Availabilites

Select

3

40

Select

4

50

Boat Search Page

Search Boat

AK1111

Search Database

Select

Boat ID

AK1111

Cust ID

2

Length

20

First Name

Emily

Last Name

Dawson

Email

emily.dawson@hotmail.com

Phone

555-2020

SQL Table Data (to show it actually gets committed to the database):

Reservations Table

	reservation_id	cust_id	boat_id	slip_number	start_date	end_date
▶	1	1	BT101A	10	2025-04-01	2025-04-15
	2	1	BT202B	11	2025-05-10	2025-05-18
	3	2	BT303C	12	2025-06-01	2025-06-12
	4	3	BT404D	13	2025-07-20	2025-07-30
	7	1	BT202B	14	1234-02-01	1234-03-05
•	NULL	NULL	NULL	NULL	NULL	NULL

Boats Table

	boat_id	cust_id	length
▶	AK1111	2	20
	AK1967	3	40
	BT101A	1	28
	BT202B	1	32
	BT303C	2	24
	BT404D	3	40
•	NULL	NULL	NULL

Slips table (1 means available, 0 means occupied)

	slip_number	availability	length
▶	1	0	20
	2	1	20
	3	1	40
	4	1	50
	10	0	30
	11	0	40
	12	0	25
	13	0	50
	14	0	30
	102	0	40
	103	0	50
•	NULL	NULL	NULL

Customers Table:

	cust_id	last_name	first_name	email	phone
▶	1	Smith	John	john.smith@gmail.com	555-1010
	2	Dawson	Emily	emily.dawson@hotmail.com	555-2020
	3	Carver	Michael	mike.carver@yahoo.com	555-3030
•	NULL	NULL	NULL	NULL	NULL